

# Estimating Player Impact in College Basketball via Regularized Adjusted Plus-Minus

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## Abstract

Regularized Adjusted Plus-Minus (RAPM) is a widely used metric for evaluating player impact in the NBA. It is a form of Adjusted Plus-Minus (APM), measuring a player's impact on a team's point differential per 100 possessions. Unlike standard box-score stats, RAPM accounts for the quality of teammates and opponents on the floor. Raw RAPM is calculated using a ridge regression model on a matrix that contains each player on the floor for a given possession, along with a vector containing the points scored. Regularization is used to limit noise from low-minute players and to reduce multicollinearity among players who often share the floor. To obtain a more stable estimate, a Bayesian technique that pushes coefficients toward a prior estimate of player impact, rather than to 0, is employed. These methods for raw and Bayesian RAPM were applied to the last three seasons of men's and women's college basketball. Despite its popularity in professional basketball, RAPM resources for college basketball are scarce, and this project was conducted to help bridge the analytics gap from professional to college. RAPM estimates can be a valuable resource for coaches seeking to recruit via the transfer portal and for analysts identifying potentially unheralded high-impact players.

## 1 Introduction

At any level of basketball, the question of "Who are the best players?" is a constant. At first glance, it seems like a simple question, but the potential avenues to answer the question are endless. Can it be answered by the eye test? In theory, sure. In practice, however, how does one manage watching every game for all 5,000+ Division 1 basketball players? Additionally, without knowledge of a team's game plans and strategies, can one effectively attribute blame to mistakes made by players? Due to these constraints, statistics are needed. For most of basketball's modern history, these statistics came from the box score. In recent decades, analysts have correctly identified that not all box scores are created equal. Open shot attempts in the flow of the offense are more efficient than contested, late-clock shots, rim assists are generally more indicative of a quality playmaker than three-point assists, and the best rim protectors have fewer opportunities to block shots because of their rim deterrence.

In its essence, the goal of basketball is for a player's team to score more points than their opponent. A player does not necessarily have to be the one who scores or assists on those points to accomplish this goal. When framed this way, we can boil the complexity of basketball down to identifying which players have the largest impact on their team's point differential. This is the basis of Regularized Adjusted Plus-Minus.

## 2 Background

The simplest form of player impact on their team's point differential is raw plus-minus. A player's raw plus-minus is simply the difference between the total number of points scored by the player's team and the total number of points scored by the opponent while that player is on the court. At first glance, this seems like a reasonable estimate of a player's impact, but it quickly falls apart. This raw plus-minus does not consider the quality of the other players on the court – both teammates and opponents. High-minute players on bad teams are almost always going to be rated as the worst players, even if that is not true. A simple strength of schedule adjustment does not suffice either; injuries and lineup patterns affect individual games differently. By definition, raw plus-minus also ignores the impact of a superstar teammate. If

on a certain team, Player A is worth +10 points per 100 possessions, and Player B is worth -2 points per 100 possessions, and Player B only plays in lineups with Player A, Player B's raw plus-minus will be very inaccurate.

The solution to this problem is a metric called Adjusted Plus-Minus (APM). This metric was developed in the early 2000s by Wayne Winston and Jeff Sagarin (Pelton, 2019). Instead of using raw plus-minus to evaluate player impact, they used regression models to account for the impact of all ten players on the floor, significantly improving the accuracy of the metric. In the early 2010s, APM was further developed by Joe Sill and Jeremias Engelmann to become RAPM, and later xRAPM (Engelmann, 2026a). RAPM follows the same process as APM, but introduces regularization – a machine learning technique meant to reduce overfitting and multicollinearity – to reduce noise and improve estimates. xRAPM introduces a prior estimate of a player's impact, allowing RAPM estimates to improve even more and converge faster.

RAPM and its variants are used by NBA front offices, and public resources such as Engelmann's xRAPM database (Engelmann, 2026b) allow fans to easily access and utilize RAPM. Unfortunately, the same level of public access is not provided for college basketball. The amount of publicly available statistical resources is much lower in college than in the NBA, and NBA play-by-play data is much easier to access than for college, making it easier for NBA fans to create their own metrics. However, recently this trend has started to change. Hoop Explorer, a website that showcases player and team stats for both men's and women's college basketball, has introduced a freely available RAPM metric (Hoopexplorer, 2026). The goal of this paper is to add to the publicly available advanced stats in college basketball, both for the men's and women's games. In addition, by spelling out the process of the creation of RAPM, readers will have the ability to implement their own versions.

### **3 Methods**

#### **3.1 General Calculation**

To construct RAPM, play-by-play data were scraped from the official NCAA website (NCAA, 2026), and Box Plus-Minus data from College Basketball Reference (LLC, 2026). For details

about the play-by-play data scraping process, please refer to our past work (Shane Faberman, 2025). For each game, raw play-by-play data was transformed into a matrix containing the result of each possession. This matrix contains 11 columns: The five players on offense, the five players on defense, and the number of points scored on each possession.

After concatenating each game matrix, the data were transformed further into  $\mathbf{X} \in \{0, 1\}^{N \times 2M}$  and  $\mathbf{y} \in \{0, 1, 2, 3, \dots\}^N$ .  $N$  represents the total number of possessions in the data set, and  $M$  is the number of players in the data set. Each player receives an offensive and defensive coefficient, hence why  $M$  is multiplied by two. In  $\mathbf{X}$ , 1 represents a player being on the floor during possession  $i$ . 0 represents a player being off the floor during possession  $i$ . From here, a ridge regression model is applied to solve for the coefficients. We use a ridge regression model because the penalty pushes the coefficients to zero. This is ideal because it helps reduce a significant amount of noise introduced by low-minute players and the multicollinearity from players on the same team. The analytical solution for a ridge regression model with parameter  $\lambda$  (determined by grid search cross-validation) is as follows:

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

The defensive coefficients for each player ( $DRAPM_p$ ) were negated to have a positive value to represent better defensive players. The total RAPM for the player  $p$  is simply the sum of the offensive coefficient ( $ORAPM_p$ ) and  $DRAPM_p$ . These values are multiplied by 100 to obtain an estimate of a player's impact per 100 possessions.

## 3.2 RAPM Flavors

### 3.2.1 Raw RAPM

The simplest form of RAPM, it directly follows the method described above. A limitation of RAPM is that it requires a lot of possessions to stabilize, preferably multiple seasons. This is particularly limiting in college; in addition to teams playing significantly fewer (shorter) games than NBA teams, many of the best players only play one or two seasons. This leads to potentially noisy and unreliable results, and there is a fair argument that the majority of college

players will never play enough possessions for the RAPM estimate to ever be reliable enough. Single-season RAPM can be extremely noisy. Results may be unreliable, and this is especially true for teams that have lost star players for significant amounts of time. In 2025-26, Caleb Wilson, JT Toppin, and Darryn Peterson missed a significant chunk of the season with various injuries. The raw RAPM numbers for these players are likely lower than they should be, and it is important to keep in mind that estimates for their teammates are less accurate.

However, despite its limitations, raw RAPM is a valuable piece of information. Since it is calculated purely from play-by-play data, it can identify players whose impact goes beyond the box score. The limitations of the box score have been discussed at length in the basketball analytics space, and raw RAPM is completely removed from the box score. To use raw RAPM most effectively, estimates should be taken with a grain of salt because of the slow convergence, and should be trusted only for players who receive consistent minutes throughout the course of the season.

### **3.2.2 Bayesian RAPM**

To combat the problem discussed above, Bayesian RAPM is introduced. This follows from Engelmann's xRAPM. Ridge regression attempts to shrink all coefficients to zero. In the context of this RAPM problem, this means that the objective function pushes all players to a neutral impact. However, we can instead shrink the coefficients to a prior estimate of a player's impact. This allows us to obtain more accurate RAPM estimates and converge to these estimates in fewer possessions. In this implementation of Bayesian RAPM, Offensive and Defensive Box Plus-Minus (OBPM and DBPM) are used as prior estimates for ORAPM and DRAPM, respectively. These values are then scaled by a function of a player's minutes played, since BPM is another metric that is noisy for low-minute players. Low-minute players have a prior of very close to zero, while high-minute players have a prior that is around the value of their BPM. In the previous description of Raw RAPM, we discussed how it was independent of the box score. By adding BPM, that is now obviously untrue. This is okay (and preferred) because, while sometimes misleading, the box score does provide information about a player that should not be fully ignored. Bayesian RAPM incorporates both a box score and a play-by-play data element,

allowing each to help correct the other's weaknesses.

In terms of implementation, we append  $2M$  rows onto the bottom of  $\mathbf{X}$  and  $\mathbf{y}$ , one for a player's offensive and defensive prior. These values are not just the per-possession scaled BPM, because the prior would get washed out very quickly due to the sheer number of possessions players play during the course of the season. Again, this value is scaled by a function of their minutes played. Bayesian RAPM provides a more comprehensive estimate of the impact of the player than raw RAPM.

### **3.2.3 Multi-season RAPM**

RAPM, as previously discussed, takes many possessions to converge. Realistically, this takes place over multiple seasons. Multi-season RAPM, both in its raw and Bayesian forms, follows the same procedure mentioned in the sections above. The prior OBPM and DBPM values are now weighted averages of each season included in the calculation.

In the NBA, multi-season RAPM is preferred when discussing the most impactful players in the league. However, in college, it is not that simple. Nikola Jokić has made an All-NBA team each of the last eight seasons. We have a roughly 20,000-minute sample size of him being an undisputed superstar. Most college careers are at most four seasons. College seasons are less than half as long as NBA seasons, and the games are eight minutes shorter. To make matters more complicated, role-change and development are accelerated in college due to the natural age curve. Once a player becomes a superstar in college, they usually either run out of eligibility or take their talents to the NBA. We don't have these massive samples of possessions where a college player is exhibiting superstar-level impact. This makes college multi-season RAPM different from NBA multi-season RAPM. It is hard to compare a freshman superstar and a senior superstar in multi-season RAPM due to the sheer difference in the number of possessions. Unlike in the NBA, where a young superstar will eventually play enough possessions to bridge this gap, a freshman phenom will likely not. In addition, the differences in role-change year-by-year are much more significant in college. Michigan State's Jeremy Fears Jr. went from a low-minute bench player to a decent starter to a superstar in subsequent seasons. That simply does not happen in the NBA.

This is not to say multi-season RAPM isn't useful in college, but it shouldn't be interpreted the same way as in the NBA. It is more a representation of the quality of the overall careers of players.

## 4 Results

RAPM data for the men's and women's 2023-24, 2024-25, and 2025-26 seasons, along with multi-season RAPM, can be found here. As of the writing of this paper, the 2025-26 season data is only partially complete, due to scraping issues. RAPM can obviously be used to rank the best players in the sport, but its potential uses expand beyond that.

### 4.1 Ranking Players

EvanMiya.com is perhaps the most trusted college basketball analytics website by fans and teams, utilized by more than 100 Division I programs (Miya, 2026). BPR, or Bayesian Performance Rating, is Miya's one-number player rating, and Bayesian RAPM mirrors many trends from BPR. Looking at All-Americans, the fifteen best players in the country, Bayesian RAPM and BPR tend to agree on a majority of them.

| Bayesian RAPM               | EvanMiya                    | AP All Americans |
|-----------------------------|-----------------------------|------------------|
| <b>Cooper Flagg</b>         | <b>Cooper Flagg</b>         | Cooper Flagg     |
| <b>Nique Clifford</b>       | <b>Johni Broome</b>         | Johni Broome     |
| <b>Kam Jones</b>            | Braden Smith                | Braden Smith     |
| <b>Johni Broome</b>         | <b>Kam Jones</b>            | Walter Clayton   |
| <b>Thomas Haugh</b>         | <b>Walter Clayton</b>       | Mark Sears       |
| <b>Joseph Tugler</b>        | <b>Dylan Cardwell</b>       | PJ Haggerty      |
| <b>Walter Clayton</b>       | <b>Thomas Haugh</b>         | Kam Jones        |
| <b>Ryan Kalkbrenner</b>     | <b>Joseph Tugler</b>        | JT Toppin        |
| Danny Wolf                  | Zakai Zeigler               | John Tonje       |
| <b>Collin Murray-Boyles</b> | Hunter Dickinson            | RJ Luis          |
| Max Shulga                  | Kadary Richmond             | LJ Cryer         |
| Henri Veesaar               | <b>Yaxel Lendeborg</b>      | Hunter Dickinson |
| <b>Dylan Cardwell</b>       | <b>Ryan Kalkbrenner</b>     | Eric Dixon       |
| <b>Yaxel Lendeborg</b>      | <b>Collin Murray-Boyles</b> | Ryan Kalkbrenner |
| Aniwaniwa Tait-Jones        | <b>Nique Clifford</b>       | Zakai Zeigler    |

Table 1: All-Americans from 2024-25, per Bayesian RAPM and EvanMiya BPR, compared to the Associated Press' official All-Americans

Bayesian RAPM and BPR agreed on nine All-Americans in 2023-24, 11 in 2024-25, and ten in 2025-26. In 2024-25, players such as Nique Clifford, Thomas Haugh, Joseph Tugler, Dylan Cardwell, Collin Murray-Boyles, and Yaxel Lendeborg were all identified as highly impactful players by both Bayesian RAPM and BPR, without receiving the same praise from the media.

## 4.2 Finding Potential Breakout Stars

In 2023-24, the Arizona Wildcats won the men's PAC-12 regular-season championship, earning a No. 2 seed in the NCAA Tournament. All-American Caleb Love headlined a talented roster that featured multiple future NBA players: Love, Pelle Larsson, Keshad Johnson, and likely Motiejus Krivas next June.

| Player          | Games | Games Started | MPG         | PPG  |
|-----------------|-------|---------------|-------------|------|
| Caleb Love      | 36    | 36            | <b>32.3</b> | 18.0 |
| Oumar Ballo     | 36    | 36            | <b>26.0</b> | 12.9 |
| Pelle Larsson   | 36    | 36            | <b>30.1</b> | 12.8 |
| Keshad Johnson  | 36    | 36            | <b>27.6</b> | 11.5 |
| Kylan Boswell   | 36    | 35            | <b>26.9</b> | 9.6  |
| Jaden Bradley   | 36    | 0             | <b>20.3</b> | 7.0  |
| KJ Lewis        | 36    | 0             | <b>18.3</b> | 6.1  |
| Motiejus Krivas | 36    | 0             | <b>12.1</b> | 5.4  |

Table 2: Minutes played for Arizona's top eight players in the 23-24 season

Sophomore guard Jaden Bradley did not start a single game for Arizona, playing just over 20 minutes per night. Despite that, he graded out as one of the most impactful players in the country according to RAPM.

| Player               | Raw RAPM    | Bayesian RAPM |
|----------------------|-------------|---------------|
| <b>Jaden Bradley</b> | <b>7.96</b> | <b>9.15</b>   |
| Pelle Larsson        | 6.64        | 8.59          |
| Caleb Love           | 5.13        | 7.29          |
| Motiejus Krivas      | 4.79        | 5.40          |
| Oumar Ballo          | 4.54        | 7.86          |
| Keshad Johnson       | 4.48        | 7.15          |
| KJ Lewis             | 3.71        | 6.24          |
| Kylan Boswell        | 3.14        | 3.91          |

Table 3: RAPM values for Arizona's top eight players in the 23-24 season

Not only was he the most impactful player on Arizona, but Bradley ranked eighth in the

country in raw RAPM. He was one spot behind Donovan Clingan, the seventh pick in the 2024 NBA draft, and one spot ahead of Alex Karaban, one of the most accomplished players in UConn's history. RAPM identified him as a star hidden in plain sight.

Two seasons later, Bradley won the Big 12 Player of the Year award after being left off the preseason All-Big 12 team, helping lead his team to the Final Four.

Due to the noise of single-season RAPM, a bench player with a high RAPM does not necessarily imply future stardom, but players with these numbers are worth keeping tabs on. For next season, a player to watch out for is Michigan's Trey McKenney. As a bench player during his freshman 2025-26 campaign, he ranked in the top 10 in the country in raw RAPM, despite playing on a team boasting three likely first-round picks in the 2026 NBA Draft.

### 4.3 Identifying Value in the Transfer Portal

In today's college basketball landscape, the transfer portal is perhaps the most important resource a team can use. RAPM can be used to identify potential values in the portal, allowing teams to sign high-impact players at lower prices and avoid high-cost, low-impact players. In the figures below, a few examples from the 2025-26 season are listed.

| Correctly Identified Underrated 2025-26 Transfers |                                                                                                    |                                                                                                    |                          |                    |                 |                                            |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------|--------------------|-----------------|--------------------------------------------|
| Player                                            | Transferred From                                                                                   | Transferred To                                                                                     | 247 Transfer Portal Rank | Bayesian RAPM Rank | Rank Difference | Result                                     |
| Henri Veesaar                                     |  Arizona        |  North Carolina | 36                       | 1                  | +35             | All-ACC 2nd team                           |
| Pryce Sandfort                                    |  Iowa           |  Nebraska       | 187                      | 23                 | +164            | All-B10 1st team                           |
| Bennett Stirtz                                    |  Drake          |  Iowa           | 12                       | 3                  | +9              | All-B10 2nd team                           |
| Dillon Mitchell                                   |  Cincinnati     |  St. John's     | 42                       | 10                 | +32             | All-BE 3rd team                            |
| Donovan Atwell                                    |  UNC Greensboro |  Texas Tech     | 170                      | 98                 | +72             | All-B12 HM                                 |
| Ja'Kobi Gillespie                                 |  Maryland       |  Tennessee      | 16                       | 4                  | +12             | All-SEC 1st team                           |
| Duke Miles                                        |  Oklahoma       |  Vanderbilt     | 202                      | 78                 | +124            | 16th in PPG, 5th in APG, 1st in SPG in SEC |
| Bryce Lindsay                                     |  James Madison  |  Villanova      | 277                      | 174                | +103            | 3rd in 3PTFG/G in BE                       |
| Aday Mara                                         |  UCLA           |  Michigan       | 49                       | 27                 | +22             | B10 DPOY, NCAA All-Tournament Team         |

Figure 1: Undervalued players in the 2025 transfer portal who outperformed expectations

| Correctly Identified Overrated 2025-26 Transfers |                                                                                                     |                                                                                               |                          |                    |                 |                             |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------|--------------------|-----------------|-----------------------------|
| Player                                           | Transferred From                                                                                    | Transferred To                                                                                | 247 Transfer Portal Rank | Bayesian RAPM Rank | Rank Difference | Result                      |
| Ian Jackson                                      |  North Carolina    |  St. John's  | 7                        | 639                | -632            | 18 MPG, 49.6% eFG%          |
| Joseon Sanon                                     |  Arizona St.       |  St. John's  | 38                       | 284                | -246            | 21 MPG, 42.9 eFG%           |
| Simeon Wilcher                                   |  St. John's        |  Texas       | 94                       | 464                | -370            | 19 MPG, 45.6% eFG%          |
| AJ Storr                                         |  Kansas            |  Mississippi | 47                       | 1041               | -994            | Benched mid-season          |
| Pop Isaacs                                       |  Creighton         |  Texas A&M   | 39                       | 256                | -217            | Started 8/33 games          |
| Jalil Bethea                                     |  Miami FL          |  Alabama     | 52                       | 1251               | -1,199          | Played just 8 MPG, 37.5 FG% |
| Kennard Davis                                    |  Southern Illinois |  BYU         | 41                       | 836                | -795            | 3PT% dropped 5.5%           |
| Wesley Yates III                                 |  USC               |  Washington  | 33                       | 112                | -79             | 45.6% eFG%, 0.82 AST/TOV    |

Figure 2: Overvalued players in the 2025 transfer portal who failed to meet expectations

Although these lists are a good starting point, more research is needed to determine if RAPM is a better predictor of future success than 247 Player Ratings. When comparing players who entered the portal in both 2024 and 2025, RAPM was more stable than 247's portal ranking (0.478 correlation compared to 0.415), and RAPM in 2024 had a higher correlation with a player's portal rank in 2025 than the correlation of a player's portal rank in 2024 and 2025. Due to the bias in this sample, this information is just a starting point for further investigation.

## 5 Conclusion

As discussed in previous sections, RAPM has its limitations. The slow convergence and lack of an adequate sample size introduce unavoidable noise, and like any statistic, it requires context and understanding to be used to its fullest. However, RAPM has been extensively tested in professional basketball and is trusted by analysts, front offices, and the media. It is the basis of more advanced one-number metrics, win probability models, lineup analysis tools, and more. Similar models and tools should be developed for college basketball.

RAPM can be further improved beyond the methods discussed in this paper. Since offensive impact is generally more important than defensive impact, the penalization parameter can differ between the offensive and defensive coefficients, helping produce more accurate estimates. Additionally, the concept of time decay can be incorporated to improve multi-season RAPM estimates. This weights the current season more than previous seasons, allowing the RAPM to preserve historical information without older, less relevant seasons dominating a player's estimate.

In terms of future work, the historical database can be expanded beyond the last three seasons, and the 2025-26 season will be updated once the scraping issue is resolved. Doing so will allow further analysis of the effectiveness and reliability of college basketball RAPM, including analyzing overarching transfer portal trends, such as whether RAPM is a useful predictor of which players succeed when transferring from mid-major or low-major schools to power conferences.

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## **A What I Learned**

Throughout this project, I learned a lot about processing and modeling with large amounts of data. The folder for this project currently takes up roughly 10 GB of storage on my computer, which is a scale of data for a project that I have not encountered before. I ran into the challenge of how to make my program able to efficiently run and process large amounts of data, which I haven't had the opportunity to experience in class projects. I also learned a lot about the process of creating a good metric, including the use of Bayesian techniques to increase a metric's accuracy. I also developed my R Shiny skills through the creation of the database.

This project improved my Python and R, as I used both for different steps in the process. Additionally, my web-scraping, data visualization, and data processing skills were improved, and from the statistical knowledge lens, I learned more about machine learning and Bayesian statistics. As a result of this project, I have a deeper understanding of all-in-one player impact metrics and a better grasp of what they represent, their strengths and weaknesses, and how to apply them. This knowledge is helpful for me, with my upcoming summer working within the field of basketball analytics with the NBA, and potentially a future career in sports analytics.